INCH-POUND

MIL-R-55342E
26 January 1990
SUPERSEDING
MIL-R-55342D
1 September 1987

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM, CHIP, ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for fixed, film, chip resistors primarily intended for incorporation into hybrid microelectronic circuits and surface mounting applications. Resistors covered by this specification will have an established reliability with life failure rates ranging from 1 percent to 0.001 percent per 1,000 hours (see 1.2.6). These failure rates are established at 60 percent confidence on the basis of life tests. The failure rate, identified by the appropriate symbol, is referred to operation at full rated voltage and rated temperature, with a permissible change in resistance of ±2 percent as the criteria for failure. Table I provides a summary of performance characteristics for these resistors. A part per million (PPM) quality system is used for documenting and reporting the average outgoing quality of resistors supplied to this specification. Statistical process control (SPC) techniques are required in the manufacturing process to minimize variation in production of resistors supplied to the requirements of this specification.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

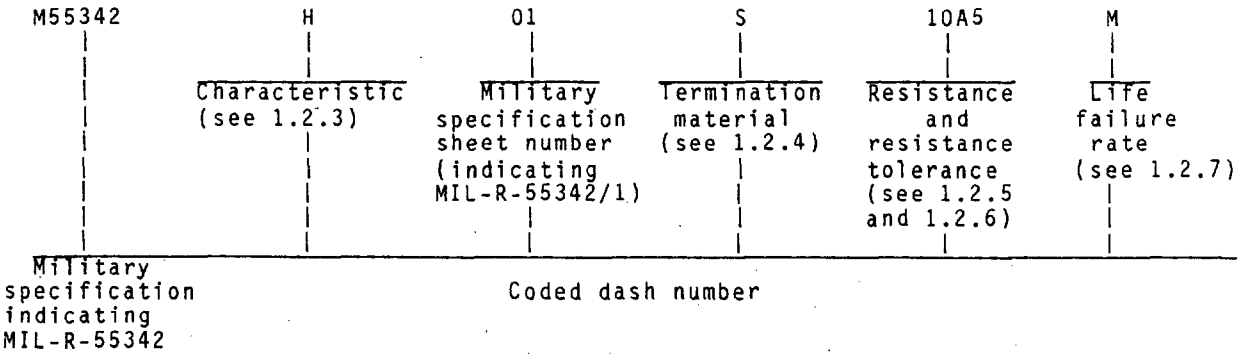
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1.2 Classification.

1.2.1 Military part number. Resistors specified herein (see 3.1) will be identified by a military part number which will consist of the basic number of the military specification sheet and a dash number. Each military specification sheet covers a different resistor style. The dash number will be coded to provide information concerning the resistor characteristic, resistance value, resistance tolerance, termination material, and life failure rate. The military part number shall be in the following form with the dash number derived as indicated:



1.2.2 Style. The style (see 3.1) is identified by the two-letter symbol "RM" followed by a four digit number; the letters identify film, fixed resistors, in a flat chip configuration.

1.2.3 Characteristic. The characteristic is identified by the single letter H, K, M, Y, or E in accordance with table I.

TABLE I. Characteristics.

Test or Condition	Units	Symbol				
		Y 1/	E	H	K	M
Resistance temperature coefficient (see 3.15)	PPM/°C	±10	±25	±50	±100	±300
Maximum ambient temperature at full wattage (see 3.6)	Temperature in °C	70	70	70	70	70
Maximum ambient temperature at zero wattage (see 3.6)	Temperature in °C	150	150	150	150	150

See footnote at end of table.

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TABLE I. Characteristics - Continued.

Test or Condition	Units	Symbol				
		Y 1/	E	H	K	M
Thermal shock (see 3.9)	Maximum percent change in resistance (0.01 ohm additional allowed for measurement error)	±.05%	±.1%	±.25%	±.5%	±.5%
Low temperature operation (see 3.10)		±.05%	±.1%	±.25%	±.25%	±.5%
Short time overload (see 3.11)		±.05%	±.1%	±.1%	±.25%	±.5%
High temperature exposure (see 3.12)		±.05%	±.1%	±.2%	±.5%	±1.0%
Resistance to bonding exposure (see 3.13)		±.05%	±.2%	±.25%	±.25%	±.25%
Moisture resistance (see 3.14)		±.1%	±.2%	±.4%	±.5%	±.5%
Life (see 3.16) 2,000 hours		±.2%	±.5%	±.5%	±.5%	±2.0%
Resistance tolerance ± percent (see table VI)	± percent	.1 1.0	.1 1.0 2.0	0.1 1.0 2.0 5.0	.1 1.0 2.0 5.0 10.0	1.0 2.0 5.0 10.0

1/ Applicable to styles RM0505, and RM1505.

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1.2.4 Termination material. The termination material designation will be in accordance with table II.

TABLE II. Termination materials.

Type	Material	Termination area	Code letters
Solderable 1/ Bondable Weldable	Gold	Wraparound 2/ One surface Bonding pads	G W 3/ P
Solderable 1/	Base metallization barrier metal, solder coated nickel	Wraparound 2/	B
	Pretinned	Wraparound One surface	R 4/ S 3/
Bondable weldable	Platinum, gold	One surface Wraparound	T U
Bondable	Platinum, silver	Wraparound 2/ One surface	K M 3/
	Palladium, silver	Wraparound One surface	C D 3/

1/ Solderable or weldable terminations will meet the solderability test. Solderable terminations will be pretinned for solder reflow operation and will meet the solderability test.

2/ On wraparound termination, the pretinning will be, as a minimum, on at least two sides and only those surfaces must meet the solderability test. Wrap-around type will be illustrated on detail specifications.

3/ See 6.4.4.

4/ Inactive for new design.

1.2.5 Resistance. The nominal resistance expressed in ohms is identified by four characters, consisting of three digits and a letter. The letter is used simultaneously as a decimal point, multiplier, and a resistance tolerance designator. For resistance values:

a. Greater than or equal to 1 ohm but less than 1,000 ohms, the letters A, D, G, J, and M are used to represent a decimal point, depending on the resistance tolerance (see tables III and IV).

b. Greater than or equal to 1,000 ohms but less than 1 megohm, the letters B, E, H, K, and N are used to represent a decimal point, depending on the resistance tolerance (see tables III and IV).

c. Greater than or equal to 1 megohm, the letters C, F, T, L, and P are used to represent a decimal point, depending upon the resistance tolerance (see tables III and IV).

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All digits preceding and following the letters (A, B, C, D, E, F, G, H, T, L, J, K, M, N, and P) of the group represent significant figures. The resistance value designators are shown in tables III and IV. Minimum and maximum resistance values shall be as specified herein (see 3.1). The standard values for every decade shall follow the sequence specified in table V of MIL-R-55342 for resistance tolerances 1.0, 2.0, 5.0, and 10.0 percent. The resistance values for tolerance 0.1 percent maybe any value within the limits specified (see 3.1), but it is preferred that values be chosen from the 0.1 percent column of table V of MIL-R-55342.

TABLE III. Designator of resistance values for resistance tolerances below ±2 percent.

Designator for .1 percent tolerance	Resistance ohms			
1A00 to 9A88 incl.	1.00	to	9.88	incl.
10A0 to 98A8 incl.	10.0	to	98.8	incl.
100A to 988A incl.	100	to	988	incl.
1B00 to 9B88 incl.	1,000	to	9,880	incl.
10B0 to 98B8 incl.	10,000	to	98,800	incl.
100B to 988B incl.	100,000	to	988,000	incl.
1C00 to 9C88 incl.	1,000,000	to	9,880,000	incl.
10C0	10,000,000			
Designator for 1 percent tolerance	**Resistance ohms**			
1D00 to 9D76 incl.	1.00	to	9.76	incl.
10D0 to 97D6 incl.	10.0	to	97.6	incl.
100D to 976D incl.	100	to	976	incl.
1E00 to 9E76 incl.	1,000	to	9,760	incl.
10E0 to 97E6 incl.	10,000	to	97,600	incl.
100E to 976E incl.	100,000	to	976,000	incl.
1F00 to 9F76 incl.	1,000,000	to	9,760,000	incl.
10F0	10,000,000			

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TABLE IV. Designation of resistance values for resistance tolerances of ±2 percent or greater.

Designator for 2 percent tolerance	Resistance ohms
1G00 to 9G10 incl.	1.00 to 9.10 incl.
10G0 to 91G0 incl.	10.0 to 91.0 incl.
100G to 910G incl.	100 to 910 incl.
1H00 to 9H10 incl.	1,000 to 9,100 incl.
10H0 to 91H0 incl.	10,000 to 91,000 incl.
100H to 910H incl.	100,000 to 910,000 incl.
1T00 to 9T10 incl.	1,000,000 to 9,100,000 incl.
10T0	10,000,000
Designator for 5 percent tolerance	**Resistance ohms**
1J00 to 9J10 incl.	1.00 to 9.10 incl.
10J0 to 91J0 incl.	10.0 to 91.0 incl.
100J to 910J incl.	100 to 910 incl.
1K00 to 9K10 incl.	1,000 to 9,100 incl.
10K0 to 91K0 incl.	10,000 to 91,000 incl.
100K to 910K incl.	100,000 to 910,000 incl.
1L00 to 9L10 incl.	1,000,000 to 9,100,000 incl.
10L0	10,000,000
Designator for 10 percent tolerance	**Resistance ohms**
1M00 to 8M20 incl.	1.00 to 8.20 incl.
10M0 to 82M0 incl.	10.0 to 82.0 incl.
100M to 820M incl.	100 to 820 incl.
1N00 to 8N20 incl.	1,000 to 8,200 incl.
10N0 to 82N0 incl.	10,000 to 82,000 incl.
100N to 820N incl.	100,000 to 820,000 incl.
1P00 to 8P20 incl.	1,000,000 to 8,200,000 incl.
10P0	10,000,000

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TABLE V. Standard resistance values for the 10 to 100 decade.

0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0	0.1	1.0	2.0 5.0	10.0
10.00	10.00	10.00	10.00	22.90	---	---	---	49.30	---	---	---
10.10	---	---	---	23.20	23.20	---	---	49.90	49.90	---	---
10.20	10.20	---	---	23.40	---	---	---	50.50	---	---	---
10.40	---	---	---	23.70	23.70	---	---	---	---	51.00	---
10.50	10.50	---	---	24.00	---	24.00	---	51.10	51.10	---	---
10.60	---	---	---	24.30	24.30	---	---	51.70	---	---	---
10.70	10.70	---	---	24.60	---	---	---	52.30	52.30	---	---
10.90	---	---	---	24.90	24.90	---	---	53.00	---	---	---
11.00	11.00	11.00	---	25.20	---	---	---	53.60	53.60	---	---
11.10	---	---	---	25.50	25.50	---	---	54.20	---	---	---
11.30	11.30	---	---	25.80	---	---	---	54.90	54.90	---	---
11.40	---	---	---	26.10	26.10	---	---	55.60	---	---	---
11.50	11.50	---	---	26.40	---	---	---	---	---	56.00	56.00
11.70	---	---	---	26.70	26.70	---	---	56.20	56.20	---	---
11.80	11.80	---	---	---	---	27.00	27.00	56.90	---	---	---
12.00	---	12.00	12.00	27.10	---	---	---	57.60	57.60	---	---
12.10	12.10	---	---	27.40	27.40	---	---	58.30	---	---	---
12.30	---	---	---	27.70	---	---	---	59.00	59.00	---	---
12.40	12.40	---	---	28.00	28.00	---	---	59.70	---	---	---
12.60	---	---	---	28.40	---	---	---	60.40	60.40	---	---
12.70	12.70	---	---	28.70	28.70	---	---	61.20	---	---	---
12.90	---	---	---	29.10	---	---	---	61.90	61.90	---	---
13.00	13.00	13.00	---	29.40	29.40	---	---	---	---	62.00	---
13.20	---	---	---	29.80	---	---	---	62.60	---	---	---
13.30	13.30	---	---	---	---	30.00	---	63.40	63.40	---	---
13.50	---	---	---	30.10	30.10	---	---	64.20	---	---	---
13.70	13.70	---	---	30.50	---	---	---	64.90	64.90	---	---
13.80	---	---	---	30.90	30.90	---	---	65.70	---	---	---
14.00	14.00	---	---	31.20	---	---	---	66.50	66.50	---	---
14.20	---	---	---	31.60	31.60	---	---	67.30	---	---	---
14.30	14.30	---	---	32.00	---	---	---	---	---	68.00	68.00
14.50	---	---	---	32.40	32.40	---	---	68.10	68.10	---	---
14.70	14.70	---	---	32.80	---	---	---	69.00	---	---	---
14.90	---	---	---	---	---	33.00	33.00	69.80	69.80	---	---
15.00	15.00	15.00	15.00	33.20	33.20	---	---	70.60	---	---	---
15.20	---	---	---	33.60	---	---	---	71.50	71.50	---	---
15.40	15.40	---	---	34.00	34.00	---	---	72.30	---	---	---
15.60	---	---	---	34.40	---	---	---	73.20	73.20	---	---
15.80	15.80	---	---	34.80	34.80	---	---	74.10	---	---	---
16.00	---	16.00	---	35.20	---	---	---	75.00	75.00	75.00	---
16.20	16.20	---	---	35.70	35.70	---	---	75.90	---	---	---
16.40	---	---	---	---	---	36.00	---	76.80	76.80	---	---
16.50	16.50	---	---	36.10	---	---	---	77.70	---	---	---
16.70	---	---	---	36.50	36.50	---	---	78.70	78.70	---	---
16.90	16.90	---	---	37.00	---	---	---	79.60	---	---	---
17.20	---	---	---	37.40	37.40	---	---	80.60	80.60	---	---
17.40	17.40	---	---	37.90	---	---	---	81.60	---	---	---
17.60	---	---	---	38.30	38.30	---	---	---	---	82.00	82.00
17.80	17.80	---	---	38.80	---	---	---	82.50	82.50	---	---
18.00	---	18.00	18.00	---	---	39.00	39.00	83.50	---	---	---
18.20	18.20	---	---	39.20	39.20	---	---	84.50	84.50	---	---
18.40	---	---	---	39.70	---	---	---	85.60	---	---	---
18.70	18.70	---	---	40.20	40.20	---	---	86.60	86.60	---	---
18.90	---	---	---	40.70	---	---	---	87.60	---	---	---
19.10	19.10	---	---	41.20	41.20	---	---	88.70	88.70	---	---
19.30	---	---	---	41.70	---	---	---	89.80	---	---	---
19.60	19.60	---	---	42.20	42.20	---	---	90.90	90.90	---	---
19.80	---	---	---	42.70	---	---	---	---	---	91.00	---
20.00	20.00	20.00	---	---	---	43.00	---	92.00	---	---	---
20.30	---	---	---	43.20	43.20	---	---	93.10	93.10	---	---
20.50	20.50	---	---	43.70	---	---	---	94.20	---	---	---
20.80	---	---	---	44.20	44.20	---	---	95.30	95.30	---	---
21.00	21.00	---	---	44.80	---	---	---	96.50			
21.30	---	---	---	45.30	45.30	---	---	97.60	97.60		
21.50	21.50	---	---	45.90	---	---	---	98.80	---		
21.80	---	---	---	46.40	46.40	---	---				
---	---	22.00	22.00	47.00	---	47.00	47.00				
22.10	22.10	---	---	47.50	47.50	---	---				
22.30	---	---	---	48.10	---	---	---				
22.60	22.60	---	---	48.70	48.70	---	---				

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1.2.6 Resistance tolerance and multipliers. The resistance tolerance and multipliers are identified by a single letter in accordance with table VI.

TABLE VI. Resistance tolerance and multipliers.

Resistance tolerance (±%)	Multiplier	Symbol
0.1	X1	A
0.1	X1,000	B
0.1	X1,000,000	C
1.0	X1	D
1.0	X1,000	E
1.0	X1,000,000	F
2.0	X1	G
2.0	X1,000	H
2.0	X1,000,000	T
5.0	X1	J
5.0	X1,000	K
5.0	X1,000,000	L
10.0	X1	M
10.0	X1,000	N
10.0	X1,000,000	P

1.2.7 Life failure rate designation. The life failure rate designation as shown in table VII is signified by a single letter (M, P, R, S), which identifies the life failure rate level for which the resistor is qualified (see 4.5 and 6.3).

TABLE VII. Life failure rate level (established at 60 percent confidence).

Failure rate level designation	Failure rate percent/1,000 hours
M	1.0
P	0.1
R	0.01
S	0.001

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

MMM-A-1931	-	Adhesive, Epoxy, Silver Filled, Conductive.
QQ-S-571	-	Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.

MILITARY

MIL-R-39032	-	Resistors, Packaging of.
MIL-I-46058	-	Insulating, Compound, Electrical (For coating printed circuit assembles).

(See supplement 1 for list of associated detail specifications).

STANDARDS

MILITARY

MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-690	-	Failure Rate Sampling Plans and Procedures.
MIL-STD-790	-	Reliability Assurance Program for Electronic Part Specifications.
MIL-STD-883	-	Test Methods and Procedures for Microelectronics.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

EIA-554	-	Assessment of outgoing nonconforming levels in Parts Per Million (PPM). (DOD adopted).
EIA-557	-	Statistical Process Control Systems (DOD adopted).

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail requirements. The individual item requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of any conflict between the requirements of this specification and the detail specifications, the latter shall govern (see 6.2).

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3.2 Qualification. Resistors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list. In addition, the manufacturer shall obtain certification from the qualifying activity that the reliability assurance requirements of 4.1.2 have been met and are being maintained.

3.3 Reliability and quality.

3.3.1 Reliability. Reliability of resistors furnished under this specification shall be established and maintained in accordance with the requirements and procedures specified in MIL-STD-790 and MIL-STD-690 with details and exceptions specified in 4.1.2, 4.4.4, 4.5, and 4.6.2.1.1.1.

3.3.2 Quality.

3.3.2.1 Statistical process control (SPC). The contractor shall implement and use statistical process control techniques in the manufacturing process for parts covered by this specification. The SPC program shall be developed and maintained in accordance with EIA-557. The SPC program shall be documented and maintained as part of the overall reliability assurance program as specified in MIL-STD-790. The implementation date for statistical process control shall be 12 months from the date of this specification. Processes for application of SPC techniques should include but are not limited to:

a. Printing of resistance elements.

b. Laser trimming.

c. Termination finish.

3.3.2.2 Quality levels. The quality of lots that have been subject to and passed the subgroup 1,100 percent screening inspection of the group A inspection shall be established and maintained in accordance with 4.6.1.2.2 and EIA-554 method B. Individual PPM defect level (i.e., PPM-2 and PPM-3) and an overall PPM defect level (i.e., PPM-5) shall be established based on the tests prescribed in the subgroup 2 tests of the group A inspections. The defect level for PPM-2 shall be less than 100 PPM. Data shall not be excluded from the appropriate PPM calculation unless specifically authorized by the qualifying activity. Guidance for exclusion of data is specified in EIA-554. The implementation date for PPM verification shall be 12 months from the date of the specification.

3.3.2.2.1 Noncompliance. The contractor shall notify the qualifying activity when the 100 PPM level is reached or exceeded for PPM-2. The contractor shall provide sufficient information to the qualifying activity documenting the causes of the problem and what corrective action is being taken. Failure to correct this problem shall be basis for removal of the affected product from the QPL.

3.4 Materials. The materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the resistors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.5 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified (see 3.1). Each resistor shall consist of a film type resistance element connected to metallized contact areas of terminations which form the electrical contacts of the resistor.

3.5.1 Passivation. Unless passivation is inherent in the device, the construction shall be passivated with a coating over the resistance element to protect the element from direct exposure to environment or handling contamination. Does not apply to trim cut.

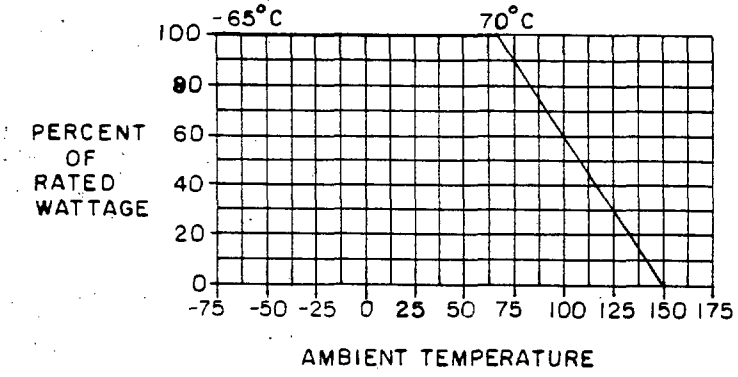
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3.5.2 Termination. Connection to the resistor element shall be made via metalized termination areas. Unless otherwise specified, both terminations (areas) shall be available on one surface of the body of the resistor chip. SEE 1.2.4 for termination codes.

3.5.2.1 Barrier metallization. The barrier metallization for the B termination (base metallization barrier metal, solder coated) shall be nickel. The metallization shall be a minimum of 50 microinches.

3.5.3 Substrate bonding. The underside of the chip resistor (the surface opposite the side containing the resistance element) may be metallized, when specified, to provide for eutectic bonding of the chip to a substrate.

3.6 Power rating. The resistors shall have a power rating based on continuous full-load operation at an ambient temperature of 70°C. For operation at temperatures in excess of 70°C the load shall be derated in accordance with figure 1.



NOTE: This curve indicates the percentage of nominal wattage to be applied at temperature higher than 70°C. This curve applies only to units mounted on a substrate (see 4.7.1.3); however, at no time shall the applied voltage exceed the maximum for each style (see 3.1).

FIGURE 1. Derating curve for high ambient temperatures.

3.7 Voltage rating. Resistors shall have a rated dc continuous working voltage or an approximate sine-wave root-mean-square (rms) continuous working voltage corresponding to the wattage (power) rating, as determined from the following formula:

$$E = \sqrt{PR}$$

Where E = Continuous rated dc or rms working voltage in volts
P = Rated wattage in watts
R = Nominal resistance in ohms

In no case shall the rated voltage be greater than the applicable maximum voltage (see 3.1).

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3.8 DC resistance. When resistors are tested as specified in 4.7.2, the dc resistance shall be within the specified tolerance of the nominal resistance (see 1.2.5).

3.9 Thermal shock. When resistors are tested as specified in 4.7.3, there shall be no evidence of mechanical damage; the change in resistance shall be within the specified initial resistance tolerance of the nominal resistance for group A tests and the change in resistance for group C tests shall not exceed ±(.25 percent +0.01 ohm) for characteristic H, ±(.50 percent +0.01 ohm) for characteristics K and M, ±(.05 percent +0.01 ohm) for characteristic Y, and ±(0.1 percent +0.01 ohm) for characteristic E.

3.10 Low temperature operation. When resistors are tested as specified in 4.7.4, there shall be no evidence of mechanical damage; the change in resistance between the initial and the final measurement at 25°C ±5°C, shall not exceed ±(0.25 percent +0.01 ohm) for characteristic H, ±(.25 percent +0.01 ohm) for characteristic K, ±(0.5 percent +0.01 ohm) for characteristic M, ±(0.05 percent +0.01 ohm) for characteristic Y, and +(0.1 percent +0.01 ohm) for characteristic E.

3.11 Short time overload. When resistors are tested as specified in 4.7.5, there shall be no evidence of arcing, burning, or charring, the change in resistance shall not exceed ±(0.1 percent +0.01 ohm) for characteristic H, ±(0.25 percent +0.01 ohm) for characteristic K, ±(0.5 percent +0.01 ohm) for characteristic M, ±(0.05 percent +0.01 ohm) for characteristic Y, and ±(0.1 percent +0.01 ohm) for characteristic E.

3.12 High temperature exposure. When resistors are tested as specified in 4.7.6, there shall be no evidence of mechanical damage and the change in resistance shall not exceed ±(0.2 percent +0.01 ohm) for characteristic H, ±(0.5 percent +0.01 ohm) for characteristic K, ±(1.0 percent +0.01 ohm) for characteristic M, ±(.05 percent +0.01 ohm) for characteristic Y, and ±(0.1 percent +0.01 ohm) for characteristic E.

3.13 Resistance to bonding exposure. When resistors are tested as specified in 4.7.7, there shall be no evidence of mechanical damage, no demetallization or leaching and the change in resistance shall not exceed ±(.25 percent +0.01 ohm) for characteristics H, K, and M, ±(0.05 percent +.01 ohm) for characteristic Y, and ±(0.2 percent +0.01 ohm) for characteristic E.

3.14 Moisture resistance. When resistors are tested as specified in 4.7.8, there shall be no evidence of mechanical damage; the change in resistance between the initial and final measurements shall not exceed ±(0.4 percent +0.01 ohm) for characteristic H, ±(0.5 percent +0.01 ohm) for characteristics K and M, ±(0.1 percent +0.01 ohm) for characteristic Y, and ±(0.2 percent +0.01 ohm) for characteristic E.

3.15 Resistance-temperature coefficient. When resistors are tested as specified in 4.7.9, the resistance-temperature coefficient, at each of the temperatures specified in table XVII referred to room ambient temperature, shall not exceed ±.0050 percent per °C (50 ppm) for characteristic H, ±.0100 percent per °C (100 ppm) for characteristic K, ±.0300, percent per °C (300 ppm) for characteristic M, ±.0010 percent per °C (10 ppm) for characteristic Y, and ±.0025 percent per °C (25 ppm) for characteristic E.

3.16 Life.

3.16.1 Qualification. When resistors are tested as specified in 4.7.10, there shall be no evidence of mechanical damage. The change in resistance between the initial measurement and any succeeding measurements up to and including 2,000 hours, shall not exceed ±(0.5 percent +0.01 ohm) for characteristics H and K, ±(2.0 percent +0.01 ohm) for characteristic M, ±(0.2 percent +0.01 ohm) for characteristic Y, and ±(0.5 percent +0.01 ohm) for characteristic E.

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3.16.2 Failure rate level determination. When resistors are tested as specified in 4.7.10, there shall be no evidence of mechanical damage to the resistance element, or enclosure. The change in resistance between the initial measurement and any of the succeeding measurements shall not exceed ±(2.0 percent +0.01 ohm) for all characteristics except Y; characteristic Y shall be ±(0.5 percent +0.01 ohm). This single failure criteria shall be applicable to all measurements during the life test for purposes of determining failure rate level qualification and is applicable as a parallel requirement with 3.16.1 to measurements made during the life test specified for qualification inspection.

3.17 Solderability (applicable to terminations B, R, G, and S). When resistors are tested as specified in 4.7.11, the immersed metallized surface shall be at least 95 percent covered with a new clean smooth coating and shall exhibit no demetallization or leaching of the terminal areas. The remaining 5 percent may contain only small pinholes or rough spots, these shall not be concentrated in one area. In case of dispute, the percentage of coverage with pinholes or rough spots shall be determined by actual measurement of these areas, as compared to the total area.

3.18 Adhesion. When resistors are tested as specifed in 4.7.12, there shall be no evidence of mechanical damage.

3.19 Visual inspection. Resistors shall be inspected to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.4, 3.5, 3.5.1 to 3.5.3, 3.19.1 to 3.19.5d, 3.21, and figures 2 through 13).

3.19.1 Conductor metallization defects. Any resistor which exhibits the following defects in the active circuit metallization shall not be acceptable. The active circuit metallization is all metal or any other material used for interconnection except metallized scribe lines, unused bonding pads and identification marking.

3.19.1.1 Conductor metallization scratches. A scratch is defined as any tearing defect that disturbs the original surface of the metallization.

a. Any scratch in metallization through which the underlying resistor material also appears to be scratched.

b. Any scratch in the interconnecting metallization which exposes resistive material or oxide anywhere along its length and reduces the width of the connecting strip to less than 50 percent of its original width.

3.19.1.2 Conductor metallization voids. A void is defined as any region in the interconnecting metallization where the underlying resistive material or oxide is visible which is not caused by a scratch. Any void in the interconnecting metallization which leaves less than 50 percent of the original width undisturbed (see figure 2).

3.19.1.3 Conductor metallization adherence. Metallization adherence is defined as any evidence of metallization lifting, peeling or blistering (see figure 6).

3.19.1.4 Conductor metallization probing. Probe marks on the interconnecting metallization other than the bonding pads that violate the scratch or void criteria.

3.19.1.5 Conductor metallization bridging (see figure 2).

a. Bridged metallization defect that reduces the distance between any two metallization areas to less than 50 percent or 0.3 mils whichever is greater.

b. Bridging between metallization and resistor pattern not intended by design that reduces the distance between the two to less than 0.1 mils.

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3.19.1.6 Conductor metallization missing.

a. More than 25 percent of metallization on a bonding pad is missing (see figure 2).

b. Any shorting bar or metal pad is missing (see figures 2 and 9).

3.19.1.7 Conductor metallization alignment. Metallization alignment is defined as any misalignment between resistor and metallization such that more than 0.5 mil of resistor on a side is exposed (see figure 3).

3.19.1.8 Conductor metallization corrosion. Metallization corrosion is defined as any evidence of localized heavy stains, metallization corrosion, discoloration or mottled metallization (see figure 8).

3.19.1.9 Excessive metallization. Excessive buildup of termination on any surface material greater than .003 inch (0.08 mm) and .008 (0.20 mm) for solderable and solderable/weldable termination respectively (see figure 5).

3.19.2 Resistor defects. Any resistor which exhibits the following defects in the active area shall not be acceptable. The active area of a resistor is that part of a resistor pattern which remains in series connection between resistor terminals and is not shorted by metallization and the resistor element is not more than two rungs past laser trimmed link.

3.19.2.1 Resistor scratches (see figure 4).

a. Any scratch within the active resistor area.

b. A scratch will normally be of considerable length, 1 mil or longer and on fine lines, will extend continuously on adjacent lines. A scratch must be indicated by either disturbance of the surface resistor or extraneous, shiny material remaining from the instrument causing the scratch.

3.19.2.2 Resistor voids (see figures 2 and 10).

a. Any void or neckdown in the active resistor area which reduces the width of the area by more than 50 percent of the original area.

b. Any void in the resistor element at the termination.

3.19.2.3 Resistor adherence. Resistor adherence is defined as any evidence of resistor film lifting, peeling or blistering (see figure 6).

3.19.2.4 Probe marks. Probe marks are defined as any probe mark on the active resistor area.

3.19.2.5 Resistor material corrosion.

a. Any evidence of localized heavy stains or corrosion of resistor material in the active resistor area (see figure 8).

b. Discoloration of tantalum based resistors due to thermal stabilization is not a cause for rejection.

3.19.2.6 Resistor bridging defects.

a. Resistor bridging caused by photolithographic defect that forms a conductive bridge across two or more adjacent active resistance areas where width of bridge path is less than one half the smallest line width in the pattern.

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b. Partial bridging defect that reduces the distance between adjacent active resistance areas to less than 0.1 mil or 50 percent of the design separation whichever is less when caused by smears, photolithographic defects or other causes except that for a partial bridge within the lines and spacing of 0.1 mil design width, visual separation (evidence at 400X) is sufficient for acceptance, and that apparent bridging when caused by laser trim slag in a nonlaser trim area is acceptable as long as the resistor material under the apparent bridge is passivated.

3.19.2.7 Laser trim faults. Any resistor which exhibits the following defects in the active area shall not be acceptable.

a. Uncut material remaining after a laser cut due to "skipping" of laser beam (see figure 7).

b. If laser cut is not straight lined, the narrowest remaining width must be equal to or greater than the width of the narrowest line within the same resistor pattern.

c. Oxide voids, cracking, or similar damage caused to the underlying material by laser beam where such damage touches active interconnects, resistor area or both.

d. For a resistor pattern with links designed for incremental laser trim, where edge of cut touches the active resistor area.

e. Any "scorching" of resistor areas by direct laser beam or spurious reflections caused by optics of the system.

f. Discoloration of tantalum based resistors in and around laser kerf is not cause for rejection.

g. For a cut in the resistor pattern which reduces a link or ladder width by more than 50 percent of the original width.

h. Usable resistance width is less than 50 percent of active resistor element (figure 10).

3.19.3 Oxide void. Any resistor which exhibits an oxide void shall not be acceptable. Any oxide void is a fault in the oxide evidenced by localized double or triple colored fringes at the edges of the defect at 100X.

a. Oxide void that allowed bridging between any two resistors, metal areas or both, not intended by design.

b. Any oxide void under metallization or resistor geometry.

3.19.4 Scribing and substrate defects. The following defects shall be cause for rejection:

a. Less than 0.5 mil oxide visible between active metallization and edge of substrate. Excluded from this are any inactive metallization lines (if applicable).

b. Any chip or crack in the active resistor, metal area, or both.

c. Any crack that exceeds 5.0 mil in length or comes closer than 1.0 mil to an active resistor area on the substrate (see figure 11).

d. Any crack in a substrate that exceeds 1.0 mil in length and points towards the active resistor area.

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e. A substrate, when scribed and broken, is left with an attached portion of an adjacent substrate which contains metallization or resistor material.

f. A crack or chip in the backside of a substrate that leaves less than 75 percent of area or, a crack or chip under a bonding pad.

g. Any crack not originating at a substrate edge is cause for rejection (see figure 11).

3.19.5 Foreign material. Conductive foreign material is defined as any substance which appears opaque under all conditions of lighting and magnification used in routine visual inspection. Material is considered attached when it cannot be removed with a nominal gas blow (of 20 psig) of dry nitrogen on air.

a. Attached conductive foreign material which reduces the separation between metallization areas to less than 0.1 mil is cause for rejection.

b. Attached conductive foreign material which reduces the separation between metallization and resistor areas to less than 0.1 mil is cause for rejection.

c. More than seven random areas having attached or unattached nonconductive on conductive foreign material that are not rejectable by the criteria of a or b shall be cause for rejection.

d. Ink residue on chip from the tester shall be cause for rejection.

3.20 Marking. Marking of individual chip resistors is not required but is permitted. A noncorrosive label containing the following information shall be inserted in each unit package. Part number designation, "JAN" marking, date code, source code, and manufacturer's production lot code (see 6.5). Date and source code shall be in accordance with MIL-STD-1285. The following is an example of the complete marking:

12345	-	CAGE
7343J	-	Date code and JAN marking
M55342H01	-	Part number
S10A5M	-	Part number
XXXXX	-	Manufacturer's production lot code

The date code shall be the date of the final assembly operation for the production lot as defined in 4.6.1.1.2. The common manufacturing record shall include the same date code as that placed on the parts covered by the record.

At the option of the manufacturer, resistors may be marked (i.e., laser, color dot, etc.). The marking shall remain legible after all tests.

3.20.1 Minimum marking legibility. When tested as specified in 4.7.1.1 the marking shall remain legible. The following is an example of minimum marking:

10A5 - Resistance, resistance tolerance and multiplier

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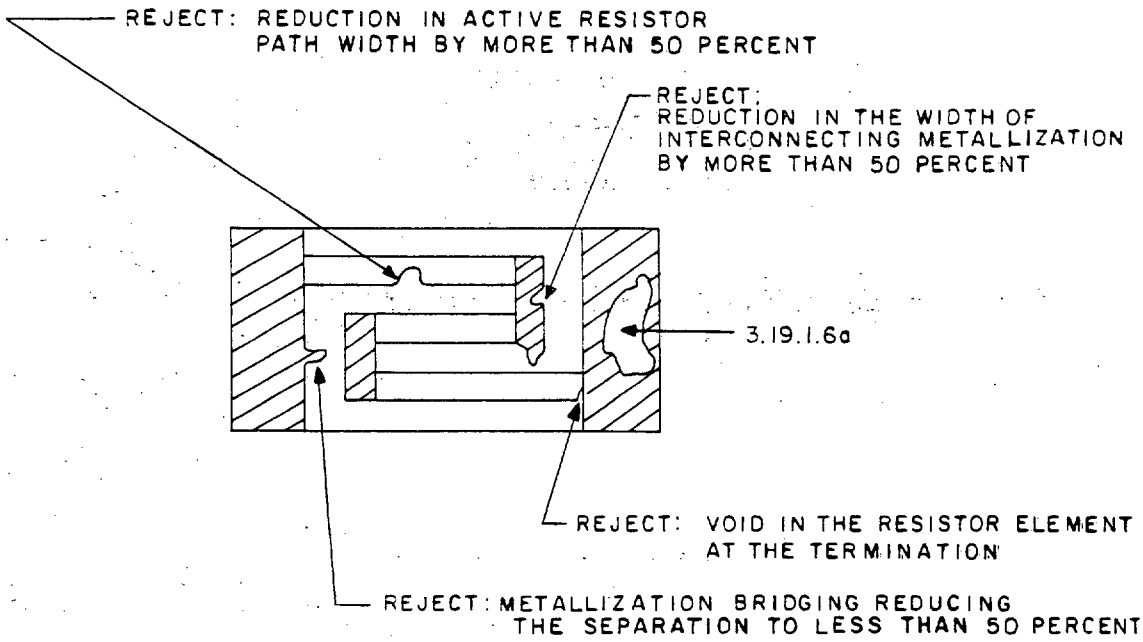
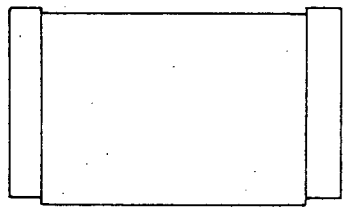


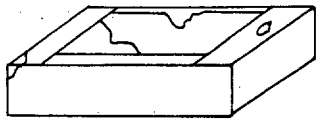
FIGURE 2. Metallization voids, bridging, missing metallization, and resistor voids.



Reject: Any misalignment between resistor and metallization.

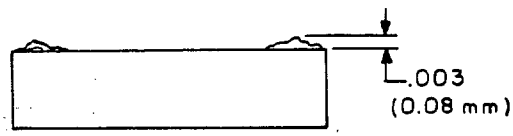
FIGURE 3. Metallization misalignment.

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REJECT: ANY METALLIZATION VOIDS OR SCRATCHES ON RESISTANCE AREA

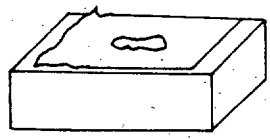
FIGURE 4. Metallization voids or scratches.



REJECT: EXCESSIVE BUILD UP OF TERMINATION ON ANY SURFACE MATERIAL GREATER THAN .003 INCH (0.08 mm) AND .008 INCH (0.20 mm) FOR SOLDERABLE AND SOLDERABLE/WELDABLE LEADS, RESPECTIVELY.

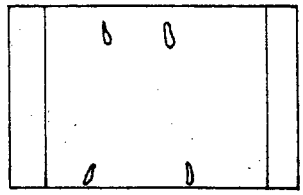
FIGURE 5. Excessive metallization.

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REJECT: ANY EVIDENCE OF PEELING, LIFTING OR BLISTERING.

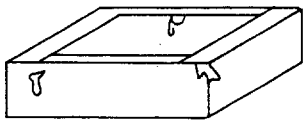
FIGURE 6. Peeling lifting or blistering metallization.



REJECT: ANY LASER TRIM FAULTS (UNCUT MATERIAL REMAINING AFTER A LASER CUT DUE TO SKIPPING OF LASER BEAM, IF LASER CUT IS NOT STRAIGHT).

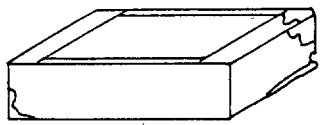
FIGURE 7. Laser trim faults.

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REJECT: METALLIZATION CORROSION (LOCALIZED STAINS, CORROSION, DISCOLORIZATION, MOTTED METALLIZATION).

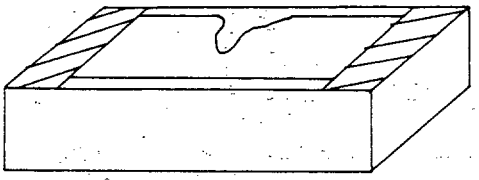
FIGURE 8. Metallization corrosion.



REJECT: MISSING BONDING PAD (ANY SHORTING BAR OR METAL PAD IS MISSING UNLESS BY DESIGN).

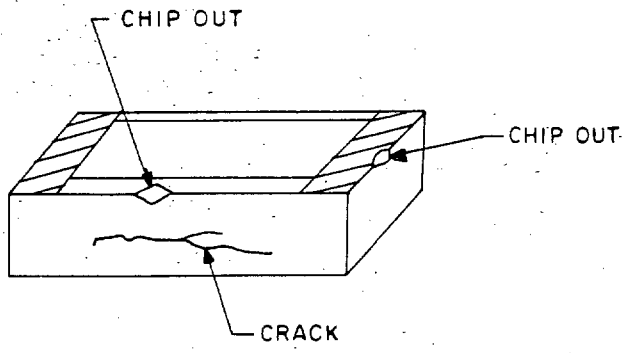
FIGURE 9. Missing metallization.

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REJECT: VOID NICK, BUBBLE OR CUT IN ACTIVE RESISTOR AREA GREATER THAN 25 PERCENT OF THE RESISTOR WIDTH EXCEPT FOR A CALIBRATION CUT AFTER CALIBRATION. THE USABLE RESISTANCE WIDTH FOR CHARACTERISTICS E, H, K AND M SHALL BE NOT LESS THAN 50 PERCENT OF THE ACTIVE RESISTOR ELEMENT

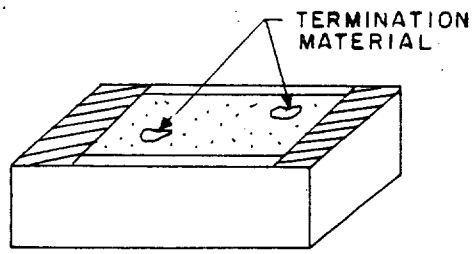
FIGURE 10. Voids, nicks, bubble or cuts.



REJECT: ANY CRACK, CHIP OUT OR VOID IN THE SUBSTRATE GREATER THAN .003 INCH (0.08 mm) IN ANY ONE DIRECTION.

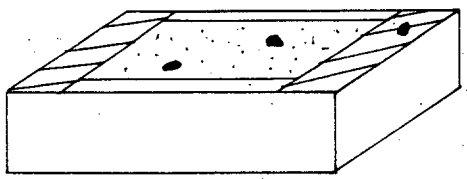
FIGURE 11. Cracks, chip outs or voids.

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REJECT: VISIBLE TERMINATION MATERIAL SPATTERED THROUGHOUT ACTIVE RESISTOR AREA.

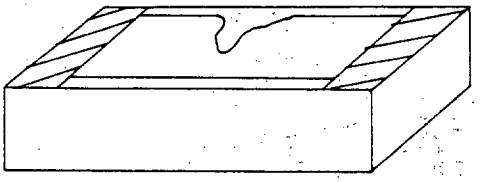
FIGURE 12. Termination material.



REJECT: VISIBLE IMBEDDED FOREIGN PARTICLES IN EITHER THE RESISTOR FILM OR TERMINATION AREA. FOREIGN PARTICLES ARE THOSE THAT ARE NOT NORMAL TO RESISTOR DESIGN OR PROCESS.

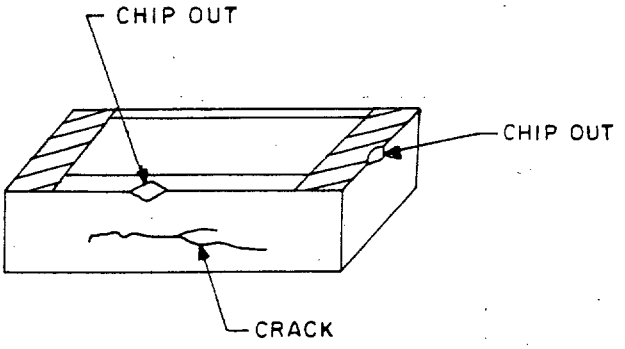
FIGURE 13. Imbedded particles.

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REJECT: VOID NICK, BUBBLE OR CUT IN ACTIVE RESISTOR AREA GREATER THAN 25 PERCENT OF THE RESISTOR WIDTH EXCEPT FOR A CALIBRATION CUT AFTER CALIBRATION. THE USABLE RESISTANCE WIDTH FOR CHARACTERISTICS E, H, K AND M SHALL BE NOT LESS THAN 50 PERCENT OF THE ACTIVE RESISTOR ELEMENT

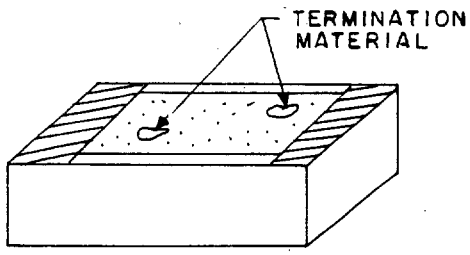
FIGURE 10. Voids, nicks, bubble or cuts.



REJECT: ANY CRACK, CHIP OUT OR VOID IN THE SUBSTRATE GREATER THAN .003 INCH (0.08 mm) IN ANY ONE DIRECTION.

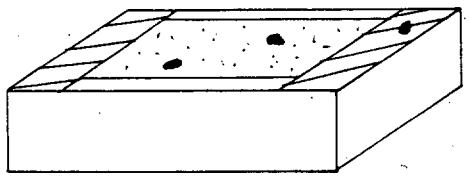
FIGURE 11. Cracks, chip outs or voids.

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REJECT: VISIBLE TERMINATION MATERIAL SPATTERED THROUGHOUT ACTIVE RESISTOR AREA.

FIGURE 12. Termination material.



REJECT: VISIBLE IMBEDDED FOREIGN PARTICLES IN EITHER THE RESISTOR FILM OR TERMINATION AREA. FOREIGN PARTICLES ARE THOSE THAT ARE NOT NORMAL TO RESISTOR DESIGN OR PROCESS.

FIGURE 13. Imbedded particles.

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3.20.2 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over, the certification marks "JAN" and "J", respectively, to indicate electrical equipment, namely resistors, capacitors, electron tubes and the like, acquired by, or manufactured for use by, or for the Government in accordance with standard government specifications. Accordingly, resistors acquired to, and meeting all of the criteria specified herein and in applicable detail specification shall bear the certification mark "JAN", except that resistors too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Resistors furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein and in applicable detail specifications shall not bear "JAN" or "J". In the event a resistor sample fails to meet the requirements of this specification and the applicable detail specification, the manufacturer shall remove the "JAN" or "J" from the sample tested and also from all resistors represented by the sample. The United States Government has obtained certificate of registration no. 504,860 for the certification mark "JAN".

3.20.3 Failure rate substitution. A manufacturer may supply to all higher failure rate levels than to which he is qualified. Parts qualified to lower failure levels with procuring activity approval, are substitutable as long as the label reflects the correct failure rate level (see table VIII). Parts that are physically marked shall not be remarked unless specified in the contract or purchase order (see 6.2).

TABLE VIII. Failure rate substitution.

Failure rate	Failure rate substitution
S (.001)	S, R, P, M
R (.01)	R, P, M
P (0.1)	P, M
M (0.0)	M

3.20.4 Resistance temperature characteristic and resistance tolerance substitution. Parts qualified to lower resistance temperature characteristics and/or lower resistance tolerance levels with procuring activity approval are substitutable for higher resistance temperature or higher tolerance levels as long as the label reflects the correct resistance temperature characteristic and/or resistance tolerance (see tables IX and X). Parts that are physically marked shall not be remarked unless specified in the contract or purchase order (see 6.2).

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TABLE IX. Resistance tolerance substitution.

Resistance tolerance	Resistance tolerance substitution
0.1	0.1, 0.5, 1.0, 2.0, 5.0, 10.0
0.5	0.5, 1.0, 2.0, 5.0, 10.0
1.0	1.0, 2.0, 5.0, 10.0
2.0	2.0, 5.0, 10.0
5.0	5.0, 10.0
10.0	10.0

TABLE X. Characteristic substitution.

Characteristic tolerance	Characteristic substitution
Y	Y, E, H
E	E, H
H	H
K	K, M
M	M

3.21 Workmanship. Resistors shall be processed in such a manner as to be uniform in quality and shall meet the requirements of 3.1, 3.4, 3.5, 3.5.1 through 3.5.3, 3.20 and 3.20.1, as applicable, and be free from other defects that will affect life, serviceability, or appearance, and shall pass the visual inspection as specified on figures 2 through 13.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Reliability assurance program . A reliability assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.1.3 Statistical process control (SPC). A SPC program shall be established and maintained in accordance with EIA-557. Evidence of such compliance shall be verified by the qualifying activity as a prerequisite for qualification and retention of qualification.

4.2 Classification of inspection. The examination and testing of resistors shall be classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Inspections conditions and precautions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the general requirements of MIL-STD-202.

4.3.2 Precautions. Adequate precautions shall be taken during inspection to prevent condensation of moisture on resistors, except during the moisture resistance temperature cycling tests.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3).

4.4.1 Sample. The number of sample units comprising a sample of resistors to be submitted for qualification inspection shall be as specified in the appendix to this specification. The sample shall be taken at random from a production run and shall be produced with equipment and procedures normally used in production, and which have been subjected to and passed the requirements of group A inspection (see 4.6.1.2). Qualification shall not be granted if group A inspection requirements are not complied with.

4.4.2 Test routine. Sample units shall be subjected to the qualification inspection specified in table XI in the order shown. All sample units with the exception of 10 sample units for group V inspection shall be subjected to the inspection of group I. The 182 sample units shall then be divided as specified in table XI for groups II, III, IV, and VI and subjected to the inspection for their particular group. Sample sizes and extent of qualification for characteristics shall be specified in the appendix to this specification.

4.4.3 Defective. Defectives in excess of those allowed in table XI shall be cause for refusal to grant qualification.

4.4.4 Failure rate level and quality level verification.

4.4.4.1 FR qualification. FR qualification shall be in accordance with the general requirements of MIL-STD-690 and the following details:

a. Procedure I: Qualification at the initial FR level. Level M (1.0 percent) of FRSP-60 shall apply. Sample units shall be subjected to the qualification inspection specified in group IV, table XI (see 4.4.2). Entire life test sample shall be continued on test to 10,000 hours as specified in 4.7.10 upon completion of the 2,000 hour qualification.

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b. Procedure II: Extension of qualification to lower FR levels. To extend qualification to the R (0.01 percent) and S (0.001 percent) FR levels, data from two or more styles of similar construction may be combined.

c. Procedure III: Maintenance of FR level qualification. Maintenance period A of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification (see 4.6.2.1).

TABLE XI. Qualification inspection.

Inspection	Requirements paragraph	Method paragraph	Number of samples	Number of defects allowed
Group I				
Visual and mechanical inspection	3.1, 3.4, 3.5, 3.5.1 through 3.5.3, 3.19, and 3.20	4.7.1	All sample units except group V	0
DC resistance	3.8	4.7.2		
Group II				
Resistance to bonding exposure	3.13	4.7.7		
Resistance temperature characteristic	3.15	4.7.9	10 highest 10 critical 10 lowest (30)	1
Low temperature operation	3.10	4.7.4		
Short time overload	3.11	4.7.5		
High temperature exposure	3.12	4.7.6		
Group III				
Resistance to bonding exposure	3.13	4.7.7	10 highest 10 critical 10 lowest (30)	1
Moisture resistance	3.14	4.7.8		1
Group IV				
Life	3.16	4.7.10	34 highest 34 critical 34 lowest (102)	1
Group V				
Solderability	3.17	4.7.11	10 any value	1
Group VI				
Adhesion	3.18	4.7.12	10 any value	1

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4.4.4.2 Quality level verification. The contractor is responsible for establishing a quality system to verify the PPM defect level of lots that are subjected to subgroup 2 tests of the group A inspections. The PPM defect level shall be based on a 6 month moving average. The contractor shall verify and report monthly the individual PPM categories (i.e., PPM-2 and PPM-3) and an overall PPM defect level (i.e., PPM-5). In the event that the contractor meets or exceeds 100 PPM for PPM-2 the qualifying activity shall take the actions specified in EIA-554 method B.

4.5 Verification of qualification. Every 6 months, the manufacturer shall compile a summary of the results of quality conformance inspections in the form of a verification of qualification report, and forward it to the qualifying activity within 30 days after the end of the reporting period as the basis of continued qualification approval. In addition, the manufacturer shall immediately notify the qualifying activity whenever the FR data indicates that the manufacturer has failed to maintain the qualified FR level or the group c inspection data indicates failure of the qualified product to meet the requirements of this specification. Continuation shall be based on evidence that over the 6 month period the following has been met:

a. Verification by the qualifying activity that the manufacturer meets the requirements of MIL-STD-790.

b. The manufacturer has not modified the design of the item.

c. The specification requirements for the item have not been amended so as to affect the character of the item.

d. Lot rejection for group A, subgroup 1, inspection does not exceed 5 percent or one lot, whichever is greater.

e. Lot rejection for group B inspection does not exceed 5 percent or one lot, whichever is greater. (Subgroup 2, group B, not included in reject rate.)

f. The requirements of group C inspection are met.

g. The records of all failure rate tests combined substantiate that the M (1.0 percent), or P (0.1 percent) FR levels have been maintained or that the manufacturer continues to meet the R (0.01 percent) or S (0.001 percent) failure rate level for which qualified, although the total component hours of testing does not, as yet, meet the requirements of 4.4.4.1c.

h. The contractor shall provide documentation to the qualifying activity pertaining to PPM calculations including numbers of part types tested, individual PPM defect categories (i.e., PPM-2 and PPM-3) and the overall PPM defect rate (PPM-5). This information shall be submitted on a detail specification basis and based on a 6-month moving average.

If group C requirements were not met and the manufacturer has taken corrective action satisfactory to the Government, the forwarding of the verification of qualification report may be delayed until within 30 days after completion of retesting of the periodic quality conformance tests. In this case, the qualifying activity shall be notified of this condition within the time that the original verification of qualification report was due. All reports shall be certified by the responsible company official and the government inspector. The qualifying activity shall be contacted for the report format.

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If group C test requires a comparison of "post test" readings with initial readings (delta measurements) the verification of qualification summary shall include the maximum and minimum delta changes for each inspection lot. For life testing delta R readings shall be reported at each interval in which readings are taken.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections. Except as specified in 4.6.2.1.3, delivery of products which have passed the groups A and B inspections shall not be delayed pending the results of the group C inspection.

4.6.1.1 Inspection and production lot.

4.6.1.1.1 Inspection lot. An inspection lot shall consist of resistors of the same style, characteristic, and termination type and manufactured under essentially the same process and conditions with the same basic materials and offered for inspection at one time.

4.6.1.1.2 Production lot. A production lot shall consist of all resistors of the same style, nominal resistance value, resistance tolerance, resistance temperature characteristic and termination material. Manufacture of all parts in the lot shall have been started, processed, assembled, and tested as a group. Lot identity shall be maintained throughout the manufacturing cycle.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table XII and shall be made on the same set of sample units, in the order shown.

4.6.1.2.1 Subgroup 1 tests. Subgroup 1 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification. Resistors that are out of resistance tolerance, or which experience a change in resistance greater than that permitted for the tests of this subgroup shall be removed from the lot. Lots having more than 5 percent total rejects, due to exceeding the specified resistance tolerance change limit shall not be furnished on contracts.

4.6.1.2.2 Subgroup 2 tests (PPM categories).

4.6.1.2.2.1 Sampling plan. Subgroup 2 tests shall be performed on an inspection lot basis. Samples subjected to subgroup 2 shall be selected in accordance with table XII based on the size of the inspection lot, (NOTE: Larger samples may be inspected by the contractor in order to calculate PPM, however rejection of the lot shall be based on 1 or more defects). In the event of 1 or more failures the lot shall be rejected. Equipment used to perform the subgroup 2 tests shall not be the same as those used in the subgroup 1 100 percent inspection tests.

4.6.1.2.2.2 PPM calculations. PPM calculations shall be based on the results of the first sample check as prescribed in 4.6.1.2.2.1. Calculations and data exclusion shall be in accordance with EIA-554. (NOTE: PPM calculations shall not use data on the second sample submission).

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4.6.1.2.3 Subgroup 3 tests. Subgroup 3 shall be performed on an inspection lot basis. Statistical sampling inspection shall be performed on an inspection lot basis. A sample of 13 parts shall then be randomly selected, if one or more defects are found, the lot shall be rescreened and defects removed. A new sample of 13 parts shall then randomly be selected. If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table XIV in the order shown, and shall be performed on sample units which have been subjected to and have passed the group A inspection.

TABLE XII. Group A inspection.

Inspection	Requirement	Method	Sampling procedure
Subgroup 1			
Thermal shock	3.9	4.7.3	100 percent
DC resistance	3.8	4.7.2	inspection
Subgroup 2 (PPM)			
(PPM-2)			
DC resistance	3.8	4.7.2	see table XIII
(PPM-3)			
Mechanical	3.5	4.7.1	
Subgroup 3			
Visual inspection	3.1, 3.4, 3.5 through 3.5.3	4.7.1	13 samples 0 failures

TABLE XIII. Sampling plans for PPM categories.

Lot size	Sample size
0 - 3200	125
3201 - 10,000	200
10,001 - 35,000	315
35,001 - 150,000	500
150,001 - 500,000	800
500,000 - UP	1250

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TABLE XIV. Group B inspection.

Inspection	Requirements paragraph	Method paragraph	Number of samples
Subgroup 1			
Resistance to bonding exposure	3.13	4.7.7	13
Resistance temperature characteristic	3.15	4.7.9	
Low temperature operation	3.10	4.7.4	
Short time overload	3.11	4.7.5	
Subgroup 2			
Adhesion	3.18	4.7.12	13
Subgroup 3			
Solderability (when applicable) 1/	3.17	4.7.11	13

1/ Destructive test.

4.6.1.3.1 Sampling plan.

4.6.1.3.1.1 Subgroup 1. A sample of 13 parts shall then be randomly selected if one or more defects are found, the lot shall be rescreened and defects removed. A new sample of 13 parts shall then be randomly be selected, if one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.3.1.2 Subgroup 2. A sample of 13 parts shall then be randomly selected, if one or more defects are found, the lot shall be rescreened and defects removed. A new sample of 13 parts shall then be randomly selected, if one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.3.1.3 Subgroup 3. A sample of 13 parts shall then be randomly selected, if one or more defects are found, the lot shall be rescreened and defects removed. A new sample of 13 parts shall then be randomly be selected, if one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.3.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.6.2 Periodic inspection.

4.6.2.1 Group C inspection. Group C inspection shall consist of the tests specified in table XV, in the order shown. Group C inspection shall be made on sample units selected from lots which have passed groups A and B inspection. Group C samples shall be representative of production. The allowable number of defectives shall be as indicated in table XV. The assigned periodic tests shall be initiated within 30 days of the specified periodic inspection.

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4.6.2.1.1 Sampling plan.

4.6.2.1.1.1 Monthly (subgroup 1). Samples shall be accumulated from each inspection lot and placed on extended life test of 4.7.10 once a month for the full 10,000 hour life test. A sufficient number of samples shall be selected from each lot by the manufacturer so that the maintenance of failure rate requirements are complied with the specified maintenance period. In any event, a minimum of 5 samples shall be selected from each lot. As far as practical, the manufacturer shall select the resistance values so that all resistance decades produced during the maintenance period are represented. The accumulated data shall be used for maintenance or extension of failure rate qualification. Subgroup 2, thermal shock test shall be performed on a sample size of 30 units with 1 defective allowed.

4.6.2.1.1.2 Quarterly. Every 3 months 60 units of each style and characteristic and termination material and of any resistance value shall be subjected to quarterly tests specified in table XV, in the order shown.

4.6.2.1.2 Disposition of samples. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.6.2.1.3 Noncompliance. If a sample fails to pass group C inspection, the contractor shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.6.3 Inspection of packaging. Sample packages and packs and the inspection of the preservation packaging, packing and marking for shipment and storage shall be in accordance with the requirements of MIL-R-39032.

4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection. Resistors shall be examined under 30X to 60X magnification, but in case of conflict 30X will be the referee power to verify that the materials, design, construction, physical dimension, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.5.1 through 3.5.3 inclusive, and 3.20).

4.7.1.1 Minimum marking. Resistors shall be coated with 0.005 inch minimum of silicon resin insulated compound, type SR of MIL-I-46058. After curing, coated resistors shall be examined for legibility under normal production room lighting by an inspector at 10X magnification.

4.7.1.2 Chip handling procedures. When specified herein, the chip resistor shall be mounted on a test substrate as described in 4.7.1.3. For those test procedures where mounting requirements are unspecified, the chip resistors may be tested unmounted using pressure type contacts.

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TABLE XV. Group C inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples	Number of defects allowed
Monthly				
Subgroup 1				
Life	3.16	4.7.10	See 4.6.2.1.1.1	See 4.6.2.1.1.1
Subgroup 2				
Thermal shock	3.9	4.7.3	30	1
Quarterly				
Subgroup 1				
Resistance to bonding exposure	3.13	4.7.7	30	
Moisture resistance	3.14	4.7.8		1
Subgroup 2				
Resistance to bonding exposure	3.13	4.7.7	30	
High temperature exposure	3.12	4.7.6		

4.7.1.3 Specified mounting. When specified in the test procedure, the chip resistor shall be mounted on a suitable test substrate which shall be a minimum of 95 percent alumina (see figure 14). The test substrate material shall be such that it shall not be the cause of, nor contribute to any failure of a chip resistor in any of the tests for which it may be used. The test substrate shall be prepared with metallized surface land areas of proper spacing so that a test substrate surface area of at least four times the resistor chip surface area is provided for each resistor chip mounted. The metallized surface land areas shall be designed in a pattern to accommodate a number of resistor chips. The metallization material shall be compatible with the bonding technique to be employed and the material used on the chip termination. The method of chip mounting for the different termination materials shall be as follows:

a. All terminations (except P and W). All chip resistors (except P and W) termination materials shall be mounted on the test substrate by soldering the chip terminations directly to the test substrate metallized land areas in the following manner:

(1) Solder paste equivalent to SN62 type "R" or "RMA" solder in accordance with QQ-S-571 shall be applied to the terminations of each chip of code letters D, T, C, U, R and B (see 1.2.4).

(2) All chips shall then be placed across the metallized land areas of the test substrate with contact between the chip terminations and substrate land areas only. The use of adhesive to keep chips in place during mounting operation is allowable.

(3) The test substrate with all chips in position shall then be placed on a suitable heat transfer unit (molten solder, hot plate, tunnel oven, etc) with the temperature maintained at 260°C for 10 ±2 seconds except chips with termination B shall be maintained at 260°C for 60 ±5 seconds.

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b. Termination material "W" and "P" (weldable). Chip resistors with weldable terminations shall be mechanically attached to the test substrate and electrically connected by thermocompression bonding flying lead interconnections between the chip termination and the test substrate metallized land areas. The interconnecting lead shall be .001 inch in diameter gold wire. The chip shall be mechanically mounted by any procedure which will not be the cause of or contribute to any failures of the chip resistor in any test.

c. At the manufacturer's option 4.7.1.3(a)1 or 4.7.1.3(b) may be used.

d. Alternative mounting for terminations D, C, T, and U. As an alternative to 4.7.1.3(a)1, chip resistors be mechanically attached to the test substrate with type Z epoxy in accordance MMM-A-1931. The epoxy shall be cured at a minimum temperature of 150°C. The minimum curing time shall be 30 minutes.

4.7.2 DC resistance (see 3.8). The dc resistance shall be measured in accordance with method 303 of MIL-STD-202. The following details and exceptions shall apply:

a. Measuring apparatus: The same measuring instrument shall be used for any one test, but not necessarily for all tests.

b. Test voltage: Measurements of resistance shall be made by using the test voltages up to the maximum values specified in table XVI. The test voltage chosen, whether it be the maximum or a lower voltage which would still provide the sensitivity required, shall be applied across the terminals of the resistor. This same voltage shall be used whenever a subsequent resistance measurement was made.

c. Temperature: The dc resistance test specified in group I of table XI shall be performed at 25°C ±2°C. For all other tests, unless otherwise specified herein, the temperature at which subsequent and final resistance measurements are made in each test shall be within ±2°C of the temperature at which the initial resistance measurement was made.

TABLE XVI. Standard dc resistance test voltages.

Nominal resistance	Maximum test voltage			
	10 to 24 Milliwatts	25 to 99 Milliwatts	100 to 225 Milliwatts	226 to 1000 Milliwatts
Ohms	Volts	Volts	Volts	Volts
5.60 to 9.88	0.2	0.3	0.7	1.0
10 to 98.8	0.3	0.3	0.3	1.0
100 to 988	1.0	1.0	1.0	3.0
1,000 to 9,880	3.0	3.0	3.0	10.0
10,000 to 98,800	10.0	10.0	10.0	30.0
100,000 or higher	30.0	30.0	30.0	100.0

4.7.3 Thermal shock (see 3.9). Resistors shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions apply:

a. Mounting: Resistors shall not be mounted. They shall be placed in containers which allow circulation of air to each resistor.

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b. Measurement before cycling:

(1) Qualification and group A tests: DC resistance shall be measured as specified in 4.7.2, except at 25°C ±5°C. The dc resistance shall be within the specified resistance tolerance of the nominal resistance value.

(2) Group C test: DC resistance shall be measured as specified in 4.7.2.

c. Test condition F.

d. Measurement after cycling:

(1) Qualification and group A tests: After stabilization at room temperature, the dc resistance shall again be measured as specified in 4.7.2, except at 25°C ±5°C.

(2) Group C test: After stabilization at room temperature, the dc resistance shall again be measured as specified in 4.7.2. Following the test, the resistors shall be examined for evidence of mechanical damage.

4.7.4 Low temperature operation (see 3.10). A dc resistance measurement shall be made as specified in 4.7.2, mounted on a substrate as specified in 4.7.1.3. The resistors shall be placed in a cold chamber maintained at -65°C +0°C, -5°C. After one hour of stabilization at this temperature, full rated continuous working voltage as specified in 3.7 shall be applied for 45 +5, -0 minutes. Fifteen +5, -0 minutes after the removal of the voltage, the resistors shall be removed from the chamber and maintained at a temperature of 25°C ±5°C for a period of approximately 24 hours; the dc resistance shall then be measured as specified in 4.7.2. Resistors shall then be examined for evidence of mechanical damage.

4.7.5 Short time overload (see 3.11). The dc resistance shall be measured as specified in 4.7.2 mounted on a substrate as specified in 4.7.1.3. A dc test potential, 2.5 times the rated continuous working voltage but not exceeding twice the maximum voltage (see 3.1) shall then be applied for 5 seconds to the resistor under the following conditions:

a. Resistors on substrate mounted so that larger space of the resistor is on a horizontal plane.

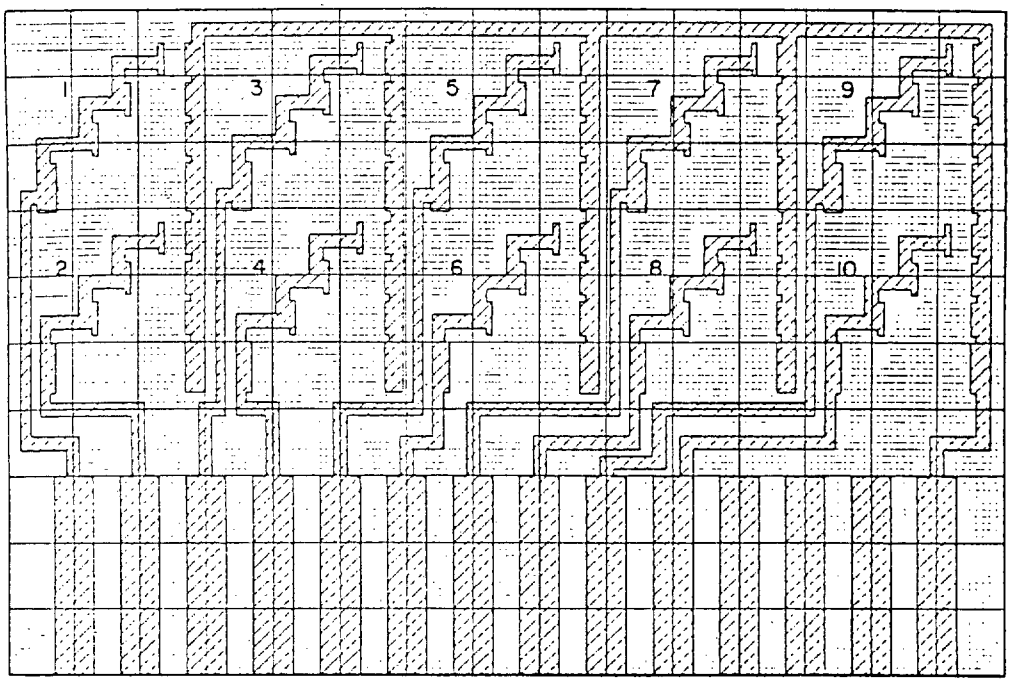
b. In still air, with no circulation other than that created by the heat of the resistors being operated.

Thirty +15, -0 minutes after removal of the test potential, the dc resistance shall again be measured as specified in 4.7.2. Resistors shall then be examined for evidence of arcing, burning, and charring.

4.7.6 High temperature exposure (see 3.12). A dc resistance shall be measured as specified in 4.7.2. The resistors shall then be subjected to a continuous temperature of 150°C ±5°C for a period of 100 ±4 hours, in a test chamber with forced air circulation. The resistors shall be stabilized at 25°C ±3°C for a period of 4 hours. At the end of the stabilization period, the dc resistance shall be measured and the change in resistance shall not exceed the value specified in 3.12.

4.7.7 Resistance to bonding exposure (see 3.13). The dc resistance of the chip resistor shall be measured as specified in 4.7.2. The chip resistor shall then be mounted on a test substrate in accordance with 4.7.1.3. The test substrate with resistors mounted shall then be stabilized at room temperature for 4 to 12 hours after which the dc resistance shall again be measured as specified in 4.7.2. The change in resistance shall not exceed the value specified in 3.13.

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NOTES:

1. Suggest substrate dimensions: 1.000 inch (25.4 mm) x 1.500 inch (38.10 mm) x .025 inch (0.64 mm) thick.
2. The contacts fit a printed circuit board connector with .100 inch (2.54 mm) in contact center.
3. Short lead wire may be soldered on and it will then fit a microcircuit connector, or test lead wires may be soldered on for direct measurement.
4. Small grid square are .010 inch (0.25 mm).

FIGURE 14. Suggested test substrate.

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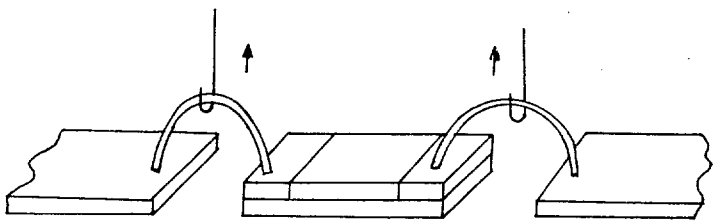


FIGURE 15. Weldable style units.

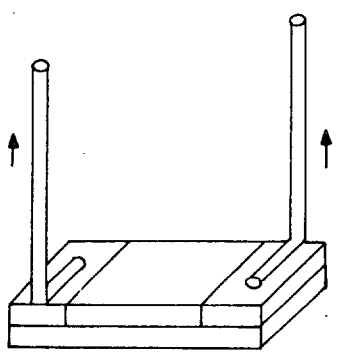


FIGURE 16. Solderable style units.

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4.7.8 Moisture resistance (see 3.14). Resistors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Chip resistor sample units shall be mounted on a test substrate as specified in 4.7.1.3.

b. Initial measurement: Immediately following the initial conditioning period, dc resistance of each chip on the test substrate shall be measured as specified in 4.7.2.

c. Polarization and loading voltage: No voltage shall be applied.

d. Subcycle: Step 7b shall not be applicable. Step 7a shall be performed during any five of the first nine cycles only.

e. Measurements at high humidity: None.

f. Final measurements: Upon completion of step 6 of the final cycle the resistors shall be removed from the chamber and air dried for 30 ±15 minutes. The dc resistance shall then be measured in accordance with 4.7.2.

g. Examination after test: Resistors shall be examined for evidence of electrical and mechanical damage.

4.7.9 Resistance temperature characteristic (see 3.15). Resistors shall be tested in accordance with method 304 of MIL-STD-202. The following details and exceptions shall apply:

a. Reference temperature: Room ambient temperature.

b. Test temperature: In accordance with table XVII.

c. Accuracy of temperature measurement: Resistors shall be maintained for 30 to 45 minutes within 1°C at each of the test temperatures listed in table XVII. This tolerance shall be maintained on the established test temperatures.

d. Mounting: Resistors shall be mounted on a substrate as specified in 4.7.1.3.

TABLE XVII. Resistance temperature characteristic.

Sequence	Temperature
	Group B acceptance inspection
	°C
1	Room temperature 1/
2	-55 ±3
3	Room temperature 1/
4	125 ±3
5	---
6	---

1/ This temperature shall be considered the reference temperature for each of the succeeding temperatures.

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4.7.10 Life (see 3.16). Resistors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting: Chip resistors sample units shall be mounted on a test substrate as specified in 4.7.1.3.

b. Test temperature and tolerance: 70°C ±5°C.

c. Initial resistance measurement of mounted resistors: Measurements may be made inside or outside the chamber.

(1) Inside chamber: When measurements are to be made inside the chamber, the initial dc resistance shall be measured after mounting at the applicable test temperature, after temperature stabilization, and within 8 hours of exposure of the resistors to the test temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

(2) Outside chamber: When measurements are to be made outside the chamber, the initial dc resistance shall be measured after mounting at room temperature. This initial measurement shall be used as the reference dc resistance for all subsequent measurements under the same condition.

d. Operating conditions: Rated dc continuous working voltage or filtered or nonfiltered full wave rectified ac voltage shall be applied intermittently, 1 hour, 30 minutes on and 30 minutes off, for the applicable number of hours (see 4.7.10f) and at the applicable test temperature. "On time" shall be three quarters of the total elapsed time. During the "on" cycle, the voltage shall be regulated and controlled to maintain ±5 percent of the rated continuous working voltage.

e. Test condition: 2,000 hours elapsed time for qualification inspection with all samples continued to 10,000 hours; 2,000 hours for 70°C power rating test. 10,000 hours for failure rate level inspection group C.

f. Measurements during test:

(1) Qualification inspection: DC resistance shall be measured at the end of the 30 minutes off periods after 250 +48, -0, 500 +48, -0, 1,000 +48, -0, and 2,000 +72, -0 hours have elapsed.

(2) Extended life testing: DC resistance shall be measured at the end of the 30 minutes off periods after 250 +48, -0, 500 +48, -0, 1,000 +48, -0, and 2,000 +72, -0, and every 2,000 +96, -0 hours thereafter until the required 10,000 hours have elapsed. Measurements shall be made as near as possible to the specified time but may be adjusted so that measurements need not be made during other than normal workdays.

(3) Measurements outside of the chamber: When measurements are made outside the chamber, resistors shall be outside of the chamber, for a minimum of 45 minutes and stabilized before measurement.

g. Examination after test: Resistors shall be examined for evidence of mechanical damage.

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4.7.11 Solderability (see 3.17) applicable to B, R, G, and S terminations. Resistors shall be tested in accordance with MIL-STD-202, method 208. Both end terminations shall be immersed completely one at a time. The following details and exceptions shall apply:

a. Apparatus:

(1) A solder pot capable of keeping the solder temperature to ±5°C of setpoint.

(2) A device to keep the specimen submerged in solder for the required time.

(3) Optical equipment capable of 30X magnification.

(4) A laboratory beaker or appropriate laboratory apparatus that is capable of:

(a) Supporting a nonmetallic device.

(b) Maintaining the test specimens at a distance of 1.5 to 2.5 inches from the surface of the boiling distilled water while, at the same time exposing them to the full flow of steam generated.

b. The requirement for standard solderable wire application shall not apply.

c. The immersion and emersion rates shall not apply.

d. The specimen shall be totally submerged for at least five seconds.

4.7.12 Adhesion (see 3.18). The resistor chips shall be prepared as specified in 4.7.12.1 or 4.7.12.2, and 4.7.12.3 as applicable. A pull of 4 grams shall be applied to the weldable style units ("W" and "P") and 15 grams to all solderable, solderable-weldable units and bondable units. The force shall be applied at a 90 degree angle to the surface of the chip (see figures 15 and 16) one lead at a time for 30 seconds each. The chip ceramic shall be held by means of tweezers or other suitable holding device. The resistor chips shall be examined for evidence of mechanical damage.

4.7.12.1 Weldable style units.

a. Method of mounting. Sample units shall be mounted on a test substrate as specified in 4.7.1.3.

b. Force. Force shall be applied by inserting a hook under the lead wire appoximately in the center of the wire as shown on figure 15.

4.7.12.2 Solderable or bondable units. A 3 inch length of solid AWG no. 26 to no. 34 wire shall be soldered to the chip terminations as shown on figure 16.

4.7.12.3 Bondable units. A 3 inch length of solid AWG no. 26 to no. 34 wire shall be epoxy bonded to the chip terminations as shown on figure 16. The epoxy shall be type 2 in accordance with MMM-A-1931.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Chip resistors are intended to be used in thin or thick film hybrid circuits where micro-circuitry is indicated and in surface mount applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable detail specification, and the complete military part number (see 3.1 and 1.2.1).

c. Termination area configuration (see 3.5.2).

d. Substrate bonding (see 3.5.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have been so listed by that date. The attention of the manufacturers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified product list is the US Army Laboratory Command; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45444.

6.4 Application notes.

6.4.1 Mounting of resistors. Resistors may be mounted individually on a substrate, usually 95 percent alumina, and connected to conductor areas by means of solder pre-forms, conductive cement or wire bonding. Users are cautioned that the substrate and chip component must have compatible temperature coefficients of expansion to reduce the strain level imposed on the solder connections. They may also be directly connected to other components on the same substrate by means of wire bonding, using the substrate as a base or carrier for the resistor.

6.4.2 Stacking of resistors. Stacking is in general discouraged since experience indicates that failure may occur due to electrolytic action in the bonding adhesive. This action occurs from the increase in temperature caused by stacking in conjunction with the operating voltage across the resistor. The bonding adhesive at the resistor chip interface chemically reacts with the active resistor material changing the noise and resistance values and eventually the resistor will open. When stacking resistors, care should be taken to compensate for the lower heat dissipation capabilities by derating the wattage rating.

6.4.3 Close tolerance resistors. Close tolerance resistors (i.e., ±0.1 percent) should be mounted on a substrate by a method which produces the least heating effect over a short time to avoid permanent changes in resistance.

6.4.4 Resistor film orientation. It is recommended that the film side of resistors be mounted up when they are placed onto printed circuit boards. This is due to the effect of thermal stressing on epoxies used to hold them in place when mounted on the bottom of printed circuit boards. Resistor films can be torn from the substrate during thermal cycling, substantially altering the value of the resistor.

6.5 Noise level test. This specification does not include a noise test, but if the user requires a noise test, it shall be performed in accordance with method 308 of MIL-STD-202.

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6.6 Electrostatic charge effects. Under relatively low humidity conditions, some types of film resistors, particularly those with small dimensions and high sheet resistivity materials, are prone to sudden significant changes in resistance (usually reductions in value) and to changes in temperature coefficient of resistance as a result of discharge of static charges built up on associated objects during handling, packaging, or shipment. Substitution of more suitable implements and materials can help minimize this problem. For example, use of cotton gloves, static eliminator devices, air humidifiers, and operator and work bench grounding systems can reduce static buildup during handling. Means of alleviating static problems during shipment include elimination of loose packaging of resistors and use of metal foil and anti-static (partly conducting) plastic packaging materials.

6.7 Terminal substitution data. For existing design and logistic support of inactivated termination R (solderable pretinned wraparound terminations). Termination B (solderable base metallization barrier metal, solder coated) shall be used as a substitute for termination R.

6.8 Subject term (key word) listing.

chip resistor

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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APPENDIX

PROCEDURES FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 Scope. This appendix details the procedure for submission of samples with related data, for qualification inspection of resistors covered by this specification. The procedure for extending qualification of the required sample to other resistors covered by this specification is also outlined herein. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this document.

30. SUBMISSION

30.1 Sample. A sample consisting of 182 sample units, 54 of the lowest, 54 of the highest, and 54 of the critical or nearest to critical value (see table XIX), and in each style and characteristic, termination material, and resistance tolerance for which qualification is sought, shall be submitted and subjected to the inspection of table XI. If no critical value is specified, the sample will be equally divided between the highest and lowest resistance values. In addition, 10 sample units, as applicable, of any value or characteristic and 10 further sample units of any resistance value shall be submitted and subjected to the tests of groups V and VI, respectively of table XI. If qualification to a closer resistance tolerance than submitted above is desired, 30 additional sample units of the closer resistance tolerance shall be subjected to the following tests(see table XVIII):

TABLE XVIII. Sample size for qualification inspection.

Inspection	Requirement paragraph	Method paragraph	Number of sample units for inspection	Number of defectives allowed
Visual and mechanical inspection	3.1, 3.4, 3.5, 3.5.1 through 3.5.3, 3.19 through 3.20	4.7.1	30 (10 highest, 10 critical, 10 lowest)	0
Thermal shock	3.9	4.7.3		
DC resistance	3.8	4.7.2		

30.1.1 Qualification to extended resistance ranges. For qualification to the extended resistance ranges of higher resistance tolerances, 54 additional samples of high or low resistance shall be subjected to the tests of groups I, II, III, and IV of table XI.

30.2 Test data. When examinations and tests are to be performed at a government laboratory, prior to submission, all sample units shall be subjected to all of the tests indicated as nondestructive in table XI. Each submission shall be accompanied by test data obtained from these tests. The performance of the destructive examinations and tests by the manufacturer on a duplicate set of sample units is encouraged, although not required. All test data shall be submitted in duplicate.

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APPENDIX

TABLE XIX. Critical resistance values for qualification ranges. 1/

Style	Critical resistance value 2/				
	Characteristics 3/				
	E	Y	H	K	M
RM0502	158		158	78.7	78.7
RM0505	63.4	63.4	63.4	31.6	31.6
RM0705	49.9		49.9	24.9	24.9
RM1005	31.6		31.6	15.8	15.8
RM1010	63.4		63.4	31.6	31.6
RM1505	15.8	15.8	15.8	10.7	10.7
RM2208	7.87		7.87	6.98	6.98
RM1206	78.7		78.7	39.2	39.2
RM2010	44.2		44.2	28.0	28.0
RM2512	78.7		78.7	39.2	39.2

1/ Maximum continuous working voltage shall be applied (see 3.1).
2/ The critical resistance value is the maximum standard resistance value which will dissipate full wattage when the maximum continuous working voltage is applied.
3/ All resistance in kilohms.

30.3 Description of items. The manufacturer shall submit a detailed description of the resistors being submitted for inspection, including materials used for the resistance element and the protective enclosure or coating. After qualification has been granted, no changes shall be made in materials, design, or construction without prior notification to the qualifying activity.

40. EXTENT OF QUALIFICATION

40.1 Extent of qualification. The resistance range included in the qualification of any one style and tolerance shall be between any two adjacent resistance values which pass the required qualification inspection. The extent of qualification between styles, characteristics, terminations, failure rate levels, and resistance tolerances shall be as specified in table XX.

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APPENDIX

TABLE XX. Extension of qualification.

Style	Will qualify style
RM0502	RM0502
RM0505	RM0505
RM0705	RM0705, RM0505
RM1005	RM1005, RM0705, RM0505
RM1010	RM1010, RM1005, RM0705, RM0505
RM1505	RM1505, RM1005, RM0705, RM0505
RM2208	RM2208
RM1206	RM1505, RM1005, RM0705, RM1206
RM2010	RM2208, RM2010
RM2512	RM0905, RM2208, RM2512

Characteristic	Will qualify characteristic	May qualify 1/ characteristic with minor additional testing
Y	E, Y, H	K, M
E	E, H	K, M
H	H, K, M	
K	K, M	---
M	M	---

Termination	Will qualify termination	May qualify 1/ termination with minor additional testing
G	G, W, P	B, R, S
W	W, P	S
P	P, W	S
B	B, R, S	G, W, P 2/
R	R, S	
S	S	
U	U, T	C, D
T	T	D
C	C, D	U, T
D	D	T

Failure rate level	Will qualify failure rate level
S	S, R, P, M
R	R, P, M
P	P, M
M	M

Resistance tolerance	Will qualify resistance tolerance
0.1	0.1, 1.0, 2.0, 5.0, 10.0
1.0	1.0, 2.0, 5.0, 10.0
2.0	2.0, 5.0, 10.0
5.0	5.0, 10.0
10.0	10.0

1/ To be determined by qualifying activity.
2/ Must contain barrier layer to qualify.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR
Navy - AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-55342E	Resistors, Fixed, Film, Chip, Established Reliability, Gen. Spec

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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